VEAB v0.4 - Scientific Assurance Controls

Release assessment | 12 November 2026 | Source: accessible v0.3 plus new additive controls

Current status

55 / 55 automated tests passed in the available source snapshot: 43 historical tests and 12 new assurance tests. The 93-test Rust/C++/Go inventory was reported by the user but its actual native worktree was unavailable for independent checking.

Control design

17 developer-constructed episodes: 11 deliberately invalid and six deliberately valid. Included geometric keep-out/obstacle violations, latency/deadline errors, unsupported goal and recovery claims, missing proposal events, timestamp ordering faults and valid edge cases. Labels are author-assigned, not independent human ground truth.

Comparator	Bad rejected	Good accepted	False accept	False reject
Terminal status	0/11	6/6	11	0
Terminal safety counter	1/11	5/6	10	1
Reconstructed safety terminal	8/11	6/6	3	0
Separate-process monitor	11/11	6/6	0	0

Scientific interpretation

The reconstructed safety-aware terminal comparator detected every one of the seven authored unsafe cases. The process monitor added three rejections for missing or contradictory evidence. It did NOT add physical unsafe-action detections beyond the strong safety-aware comparator. These counts are software-fixture regression outcomes, not generalisable benchmark findings.

Critical negative finding

The monitor accepts a coherently forged but internally consistent trajectory. An independent code path does not establish trustworthy source provenance; hash chaining detects alteration after commitment, not necessarily falsification before capture.

Prospective study

The hypothesis remains unproven. The primary endpoint must compare independently adjudicated unsafe task completions at a fixed false-rejection threshold, under matched scenes, sensors, controls and fault conditions. Analyse independently authored scenes, not repeated deterministic seeds. A normal-approximation illustration of a 0.20 effect and scene SD of 0.40 estimates 32 scenes; those assumptions must be justified by independent pilot data before preregistration.

Scope and exclusions

Current output: standalone read-only monitor, generated synthetic controls, comparator report, blind-review template and Cohen kappa calculation, scene-level power planning, integration instructions and automated tests. No native Isaac Lab/MuJoCo/Gazebo/ROS 2 trial, hardware test, independently blinded human review, sensor attestation, physical safety certification or SOTA claim was established.